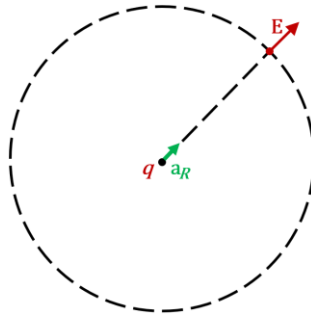


1. [25 pts] [Understanding of Coulomb's Law, Basic Concept of Divergence or Gauss's Law, Electrical Flux Density, and Electric potential] As shown below Figure, a point charge q is at the origin in free space with permittivity, ϵ_0 .



(a) [5 pts] Describe succinctly Gauss's Law and write the equations with both volume integration and surface integration forms if the charge is induced by volume charge density, ρ_v .

Gauss's Law states: "The total outward flux of the electric field intensity, $\vec{E}$ over any closed surface in free charge space is equal to the total charge enclosed in the surface divided by ϵ_0 "

$$\int_{vol} \nabla \cdot \vec{E}(\rho_v) dv = \frac{1}{\epsilon_0} \int_{vol} \rho_v dv \rightarrow \oint_S \vec{E} dS = \frac{Q}{\epsilon_0} = \frac{q}{\epsilon_0}$$

(b) [5 pts] Find electric field intensity, $\vec{E}$ considering Figure above. Write all the expansions to obtain $\vec{E}$.

$$\oint_S \vec{E} dS = \frac{q}{\epsilon_0} \rightarrow \oint_S \vec{E} dS = E_R(\hat{a}_R) \cdot (\hat{a}_R) 4\pi R^2 = \frac{Q}{\epsilon_0}$$

$$E_R = \frac{Q}{4\pi\epsilon_0 R^2}, \text{ so, } E = \hat{a}_R \frac{Q}{4\pi\epsilon_0 R^2}$$

(c) [5 pts] Find electrical flux density $\vec{D}$.

$$\vec{D} = \epsilon_0 \vec{E} = \epsilon_0 \hat{a}_R \frac{Q(q)}{4\pi\epsilon_0 R^2} = \frac{q}{4\pi * R^2} \hat{a}_R$$

(d) [5 pts] If another charge q_a is at the point A, which has a distance of R_a from the origin with the direction of $\hat{a}_R$ ($R_a > R$), what is the force acting on a charge q , arising from charge, q_a ?

$$\vec{F}_{q_a * q} = \hat{a}_R q_a \vec{E}_R = \hat{a}_R \frac{q q_a}{4\pi\epsilon_0 R_a^2}$$

(e) [5 pts] Find the electrical potential, V , of a distance R from a charge Q referred to that at infinity.

$$V_R = - \int_{\infty}^R \vec{E} \cdot d\vec{l} = - \int_{\infty}^R \hat{a}_R \frac{Q}{4\pi\epsilon_0 R^2} \cdot \hat{a}_R dR = - \frac{Q}{4\pi\epsilon_0} \left(-\frac{1}{R} \right) \Big|_{\infty}^R = \frac{Q}{4\pi\epsilon_0 R} \left[\frac{1}{R} - \frac{1}{\infty} \right] = \frac{Q}{4\pi\epsilon_0 R}$$

2. [15 pts] [Applications of Gauss's Law to find $\vec{D}$, $\vec{E}$, and $\vec{V}$]. Answer below questions considering "Gauss's Law".

(a) [10 pts] For a line charge, $\rho_l = \frac{10^{-9}}{2} \left[\frac{C}{m} \right]$, on the z - axis (in the free space), find $\vec{D}$, $\vec{E}_A$, $\vec{E}_B$, and V_{AB} where A is $(2m, \frac{\pi}{2}, 0)$ and B is $(4m, \pi, 5m)$.

According to Gauss's Law,

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0} = \frac{\rho_l L}{\epsilon_0} \rightarrow 2\pi r L E_r = \frac{\rho_l L}{\epsilon_0}$$

$$\vec{E} = \hat{a}_R \frac{\rho_l}{2\pi\epsilon_0 r}$$

$$(1) \vec{D} = \epsilon_0 \vec{E} \rightarrow \vec{D} = \epsilon_0 \hat{a}_R \frac{\rho_l}{2\pi\epsilon_0 r} = \hat{a}_R \frac{\rho_l}{2\pi r}$$

For 2 and 3, Angle and Z height are irrelevant because the line charge is at the z-axis

$$(2) \vec{E}_A = \hat{a}_R \frac{\rho_l}{2\pi\epsilon_0 r}; \quad A (2m, \frac{\pi}{2}, 0), r=2; \quad \hat{a}_R \frac{\left(\frac{10^{-9}}{2}\right)}{\left(2\pi * 2 * \frac{10^{-9}}{36\pi}\right)} = 4.5 \hat{a}_R \left[\frac{V}{m} \right]$$

$$(3) \vec{E}_B = \hat{a}_R \frac{\rho_l}{2\pi\epsilon_0 r}; \quad B (4m, \pi, 5m), r=4; \quad \hat{a}_R \frac{\left(\frac{10^{-9}}{2}\right)}{\left(2\pi * 4 * \frac{10^{-9}}{36\pi}\right)} = 2.25 \hat{a}_R \left[\frac{V}{m} \right]$$

$$(4) V_{AB} = V_B - V_A = - \int_B^A \frac{\left(\frac{10^{-9}}{2}\right)}{\left(2\pi * r * \frac{10^{-9}}{36\pi}\right)} dR = 9 * \ln(2) \cong 6.24V$$

(b) [5 pts] Given the field, $\vec{E} = \left(-\frac{16}{r^2}\right) \hat{a}_r \left[\frac{V}{m} \right]$ in spherical coordinates, find $\vec{D}_B$, and the electrical potential V of point A $(2m, \pi, \pi/2)$ with respect to $(4m, 0, \pi)$.

$$(1) \vec{D} = \epsilon_0 \vec{E} = -\frac{16}{R^2 \epsilon_0} \hat{a}_R \left[\frac{C}{m} \right] = \frac{16}{16^2 \left(\frac{10^{-9}}{36\pi}\right)} \hat{a}_R = 2.25\pi * 10^9 \hat{a}_R \left[\frac{C}{m} \right]$$

$$(2) V_{AB} = V_B - V_A = - \int_B^A -\frac{16}{R^2} dR = -4V$$

3. [20 pts] [Understanding of Dielectrics, the relationship between $\vec{D}$, $\vec{E}$, Polarization $\vec{P}$, electrical susceptibility, relative (dielectric) permittivity, and absolute permittivity]

(a) [10 pts] Find the magnitudes of $\vec{D}$, $\vec{P}$, and ϵ_r for a dielectric material in which $E = 0.15 \text{ MV/m}$ and $\chi_e = 4.25$

$$\chi_e = \epsilon_r - 1; \epsilon_r = 5.25$$

$$(1) \vec{D} = \epsilon_0 \epsilon_r \vec{E} = \frac{10^{-9}}{36\pi} * 5.25 * 0.15 * 10^6 = 6.96 * 10^{-6} \left[\frac{C}{m^2} \right]$$

$$(2) \vec{P} = \epsilon_0 \chi_e \vec{E} = \frac{10^{-9}}{(36\pi)} * 4.25 * 0.15 * 10^6 = 5.63 * 10^{-6} \left[\frac{C}{m^2} \right]$$

$$(3) \epsilon_r = \chi_e + 1 = 4.25 + 1 = 5.25$$

(b) [10 pts] Determine the value of $\vec{E}$ in a material for which the electrical susceptibility, $\chi_e = 3.5$ and $\vec{P} = 2.3 \times 10^{-7} \hat{a} [C/m^2]$

$$\vec{P} = \epsilon_0 \chi_e \vec{E} \rightarrow \vec{E} = \frac{\vec{P}}{\epsilon_0 \chi_e} = \frac{(2.3 \times 10^{-7})}{\left(\left(\frac{10^{-9}}{(36\pi)} \right) 3.5 \right)} \hat{a} \left[\frac{C}{m^2} \right] = 7.42 * 10^3 \left[\frac{V}{m} \right]$$

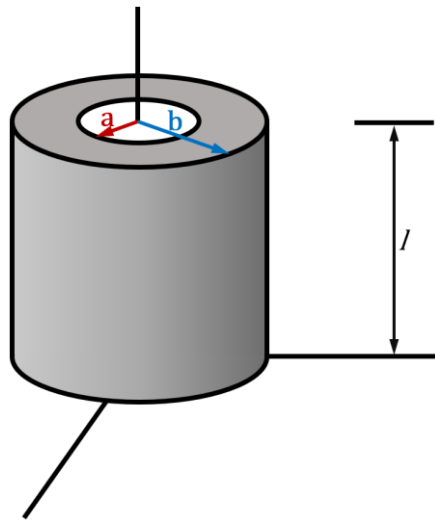
4. [20 pts] [Understanding how to find capacitance from different types of capacitors, and understanding of Multi-Dielectric Capacitors]

(a) [10 pts] A coaxial capacitor of length L , where the inner conductor has radius a and the outer has radius b as below Figure. The space between the inner and outer conductors with a dielectric permittivity, ϵ . Find:

(i) Electrical Potential, V_{ba}

(ii) Capacitance C

(iii) The stored energy in the electric field



(i) At $r=a$, if ρ_s is surface charge at the surface of inner conductor

$$\vec{B} = \rho_s \frac{a}{r} \hat{a}_r \rightarrow \vec{E} = \frac{\rho_s a}{\epsilon r} \hat{a}_r, \text{ Therefore, } V_{BA} = V_A - V_B = - \int_A^B \left(\frac{\rho_s a}{\epsilon r} \right) \hat{a}_r \cdot \hat{a}_r dR = \frac{\rho_s a}{\epsilon} \ln \left(\frac{a}{b} \right)$$

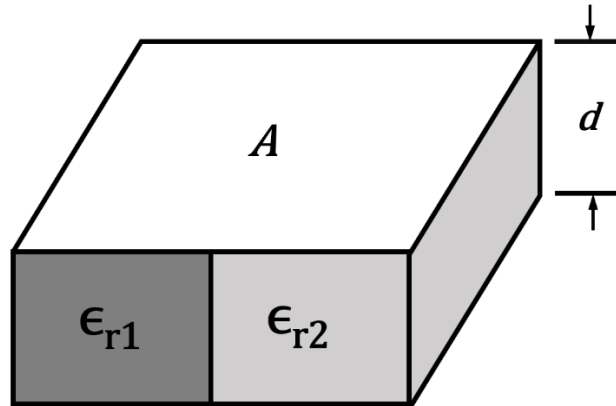
(ii) $Q = CV \rightarrow Q = \rho_0 A = \rho_s * 2\pi * a * L$

$$V_{BA} = \frac{\rho_s a}{\epsilon} \ln \left(\frac{a}{b} \right), \quad C = \frac{Q}{V} = \frac{\rho_s 2\pi a L}{\left(\frac{\rho_s a}{\epsilon} \ln \left(\frac{a}{b} \right) \right)} = \frac{2\pi \epsilon L}{\ln \left(\frac{a}{b} \right)}$$

(iii) $W_e = \frac{1}{2} \int \vec{D} \cdot \vec{E} dv = \frac{1}{2} QV = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$

$$W_e = \frac{1}{2} QV = \frac{1}{2} (\rho_s 2\pi L) \cdot \left(\frac{\rho_s a}{\epsilon} \right) \ln \left(\frac{a}{b} \right) = \frac{1}{2} \frac{\rho_s^2 a^2 2\pi L}{\epsilon} \ln \left(\frac{a}{b} \right)$$

(b) [10 pts] Find the capacitance of a parallel-plate capacitor containing two dielectrics, $\epsilon_{r1} = 1.5$ and $\epsilon_{r2} = 3.5$, each comprising one-half the volume, as shown in the Figure below. Here, $A = 2m^2$ and $d = 10^{-3}m$. In addition, repeat if the two dielectrics each occupy one-half of the space between the plates but the interface is parallel to the plates.



$$\begin{aligned}
 \text{(i)} \quad C_{total} &= C_1 + C_2 \\
 C_1 &= \frac{\epsilon_0 \epsilon_{r1} A_1}{d_2}; C_2 = \frac{\epsilon_0 \epsilon_{r2} A_2}{d_1}; A_1 = A_2 = 0.5A = 1m^2 \\
 C_1 &= \frac{\epsilon_0 \epsilon_{r1} A_1}{d} = \frac{\left(\frac{10^{-9}}{36\pi}\right)(1.5) * 1}{10^{-3}} = 1.326 * 10^{-8}F \\
 C_2 &= \frac{\epsilon_0 \epsilon_{r2} A_2}{d} = \frac{\left(\frac{10^{-9}}{36\pi}\right)(3.5) * 1}{10^{-3}} = 3.0946 * 10^{-8}F \\
 C_{total} &= 1.326 * 10^{-8}F + 3.0946 * 10^{-8}F = 4.421 * 10^{-8}F
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad C_{total} &= \frac{1}{\frac{1}{C_1} + \frac{1}{C_2}}; d_1 = d_2 = 0.5d = 0.5 * 10^{-3}; A_1 = A_2 = A = 2m^2 \\
 C_1 &= \frac{\epsilon_0 \epsilon_{r1} A_1}{d} = \frac{\left(\frac{10^{-9}}{36\pi}\right)(1.5) * 2}{0.5 * 10^{-3}} = 5.305 * 10^{-8}F \\
 C_2 &= \frac{\epsilon_0 \epsilon_{r2} A_2}{d} = \frac{\left(\frac{10^{-9}}{36\pi}\right)(3.5) * 2}{0.5 * 10^{-3}} = 1.2787 * 10^{-7}F \\
 C_{total} &= \frac{1}{\frac{1}{5.305 * 10^{-8}} + \frac{1}{1.2787 * 10^{-7}}} = 3.7136 * 10^{-8}F
 \end{aligned}$$

5. [20 pts] [Electrostatic Energy or Stored Energy by Electrostatic Field]

(a) [10 pts] Given the potential function $V = 2x + 4y$ [V] in free space, find the stored energy in a 1 m^3 volume centered at the origin

$$\vec{E} = -\nabla V = -\left(\frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z\right) = -2\hat{a}_x - 4\hat{a}_y \left[\frac{V}{m}\right]$$

$$V = 2x + 4y$$

$$|\vec{E}| = \sqrt{(-2)^2 + (-4)^2} = \sqrt{20} \left[\frac{V}{m}\right]$$

$$w_e = \frac{1}{2}\epsilon_0 E^2 = \frac{1}{2}\left(\frac{10^{-9}}{(36\pi)}\right) * (\sqrt{20})^2 = \frac{10^{-8}}{(36\pi)} \left[\frac{J}{m^3}\right]$$

$$W_e = \frac{1}{2}\epsilon_0 E^2 * V = \frac{10^{-8}}{(36\pi)} * 1 = \frac{10^{-8}}{(36\pi)} [J]$$

(b) [10 pts] The electric field between two concentric cylindrical conductors at $r = 0.01 \text{ m}$ and $r = 0.05 \text{ m}$ is given by $\vec{E} = \left(\frac{10^5}{r}\right)\hat{a}_r \left[\frac{V}{m}\right]$, fringing neglected. Find the energy stored in a 0.5 m length. Assume free space.

$$\begin{aligned} W_e &= \frac{1}{2} \int_{vol} \epsilon_0 E^2 dV = \frac{\epsilon_0}{2} \int_{vol} E^2 dV = \frac{\epsilon_0}{2} \int_0^L \int_0^{2\pi} \int_{r_1}^{r_0} \left(\frac{10^5}{r}\right)^2 r dr d\phi dz \\ &= \frac{\frac{10^{-9}}{36\pi}}{2} \int_0^{0.5} \int_0^{2\pi} \int_{0.01}^{0.05} \left(\frac{10^5}{r}\right)^2 r * dr dy dz = 0.2235 [J] \end{aligned}$$